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Improved Quantum Approximate Optimization Algorithm based on Conditional Value-at-Risk for Portfolio Optimization

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Abstract

This paper presents an improved Quantum Approximate Optimization Algorithm variant based on Conditional Value-at-Risk for addressing portfolio optimization problems. Portfolio optimization is a NP-hard combinatorial problem that aims to select an optimal set of assets and their quantities to balance risk against expected return. The proposed approach uses the QAOA to find the optimal asset combination that maximizes returns while minimizing risk, with a focus on the tail end of the loss distribution. An enhanced QAOA ansatz introduced that offers a balance between optimization quality and circuit depth, leading to faster convergence and higher probabilities of obtaining optimal solutions. Experiments are conducted using historical stock data from Nasdaq, optimizing portfolios with varying numbers of stocks. In the case of 16 stocks, our method achieves optimal cost values with merely 35 iterations, whereas the standard QAOA requires 700 iterations. Our method outperforms other approaches, particularly as the size of the problem increases.

Keywords: Quantum Approximate Optimization Algorithm, Portfolio Optimization , Quantum Computing, Conditional Value-at-Risk

1 Introduction

Mean-variance portfolio optimization (MVPO)[1], a cornerstone issue in Modern Portfolio Theory (MPT) as articulated by Markowitz, is focused on identifying an asset

portfolio that promises the highest anticipated financial returns coupled with the lowest level of variance, contingent upon a given risk tolerance. In practical scenarios of MVPO, transactions are limited to discrete quantities of assets, rendering MVPO a quadratic unconstrained binary optimization (QUBO) problem, which is widely recognized as NP-hard [2]. Despite the existence of numerous classical heuristic and metaheuristic methods for approximate QUBO problem-solving, [3–9] it was Farhi who initially proposed two quantum metaheuristic strategies: Quantum Annealing (QA) [10] and the Quantum Approximate Optimization Algorithm (QAOA) [11]. These algorithms are currently at the center of significant discourse, with the debate surrounding their potential to outperform classical methods, especially when implemented on near-future noisy intermediate-scale quantum (NISQ) devices.

Indeed, QA is tailored for specialized quantum annealer devices, which have now expanded to approximately 1000 qubits—albeit in a non-universal capacity [12]. In contrast, QAOA is adaptable to universal gate-model quantum computers, which, at the time of this writing, have not yet achieved the same level of scalability. This discrepancy in scale has led to a focus on QA for MVPO in various studies[13–18], while the application of QAOA to MVPO has been less explored [19–22]. Consequently, until this current research, a systematic comparison of the QAOA approach has not been conducted.

In this study, we concentrate on examining the QAOA and its capacity to address the discrete MVPO problem. To extend the findings of prior research [20, 21] , it is essential to deeply understand every component of the QAOA's hybrid quantum-classical framework. The QAOA is part of a broader category of variational algorithms [21, 23], where a quantum circuit's parameters are refined by a classical optimization process to maximize or minimize an objective function, typically the expected value of a specific observable, which is measured on a Quantum Processing Unit.

Inspired by [24], our approach previously uses the QAOA to find the optimal asset combination that maximizes returns while minimizing risk, with a focus on the tail end of the loss distribution as estimated by Conditional Value at Risk(CVaR)[24]. In addition, we introduces an improved QAOA ansatz that offers a balance between circuit depth and optimization quality, leading to faster convergence and higher probabilities of obtaining optimal solutions.[25]

The structure of the paper is outlined as follows: Section 2 explains the conversion of the portfolio optimization issue into an Ising model. Section 3 outlines our proposed approach. Section 4 presents the careful planning and implementation of the experiments conducted to achieve the results. Section 5, ultimately, analyzes the outcomes of our method, highlighting its successes and concludes with an overall summary.

2 Portfolio Optimization and its Formulation

The process of Portfolio Optimization involves five steps which is illustrated in Fig 1. We will discuss the details of these five steps in section 2 and section 3.

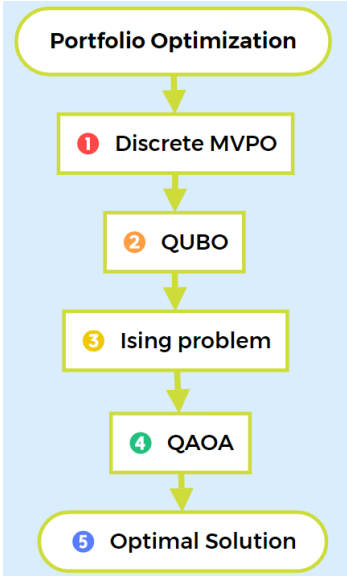

Fig. 1 Procedure of Portfolio Optimization

2.1 Portfolio Optimization

MVPO aims to optimize the Markowitz objective function $\mathcal{C}(\boldsymbol{w})$[26, 27]

$$\mathcal{C}(\boldsymbol{w}) = (1-\lambda)\boldsymbol{w}\Sigma\boldsymbol{w}^T - \lambda\boldsymbol{\mu}\cdot\boldsymbol{w} \tag{1}$$

and its minimization concerning the vector of portfolio weights $\boldsymbol{w}$ for a collection of n assets, while adhering to budgetary restriction B.

$$\min\{\mathcal{C}(\boldsymbol{w})|\sum_{i=0}^{n-1} w_i = B\}$$

to discover the ideal portfolio weights vector $\boldsymbol{w}^*$: $\boldsymbol{w}^* = argmin[\mathcal{C}(\boldsymbol{w})]$. In equation 1, $\lambda \in [0,1]$ represents the investor's risk tolerance, Σ denotes the $n\times n$ covariance matrix, and μ signifies the vector of expected returns for n assets. With historical market data provided for μ and Σ, the investor's risk tolerance, λ. A portfolio that achieves $\boldsymbol{w}^*$ is considered to be on the efficient frontier, as referenced in [28]. Traditionally, $-1 \leq w_i \leq 1$, and $B = 1$. This setup constitutes a continuous optimization issue. However, it does not reflect the practical situation where trading occurs in discrete quantities. Section 2.2 will introduce a straightforward and instructive discrete version of the MVPO .

2.2 MVPO to a QUBO Problem

To streamline, we opt for a binary weight system with $w_i \in \{0,1\}$ and $B \in \mathbb{Z}_0^+$, $0 \leq B \leq n$, establishing a clear buy/hold (1/0) strategy. Assuming uniqueness in the elements of μ and the non-redundant elements of Σ , this ensures a non-degenerate objective function. For clarity, we redefine portfolio weights for the buy/hold MVPO

as $w_i \to x_i \in \{0,1\}$. This way, it is clear that $\mathcal{C}(\boldsymbol{x})$ takes on the QUBO form

$$\mathcal{C}(\boldsymbol{x}) = \sum_{i=1}^{n} b_i x_i + \sum_{i,j=1}^{n} A_{ij} x_i x_j \tag{2}$$

for QUBO matrix elements A_{ij} . $A \in \mathbb{R}^{n \times n}$. The objective of a QUBO problem is to identify the optimal vector. The significance of the QUBO form in Equation 2 lies in its ability to express $\mathcal{C}(\boldsymbol{x})$ using spin variables $z_i \in \{-1, 1\}$, and the associated spin vector $\boldsymbol{z}$. By applying the substitution $x_i = (1 - z_i)/2$, the equation is converted into a quadratic Ising model.

$$\mathcal{C}(\boldsymbol{z}) = \sum_{i=1}^{n} d_i z_i + \sum_{i,j=1}^{n} Q_{ij} z_i z_j \tag{3}$$

The goal is $\boldsymbol{z}^* = arg \min_{\boldsymbol{x} \in \{-1,+1\}^n} \mathcal{C}(\boldsymbol{z})$. The QUBO formulation involves quadratic coupling matrix elements Q_{ij} and linear coefficients d_i. It is essential to reconstruct the value of $\mathcal{C}(\boldsymbol{z})$. As detailed in Section 3, the transformation to spin variables facilitates the encoding of $\mathcal{C}(\boldsymbol{z})$ into a Hamiltonian that is diagonal within the computational basis.

3 Improved QAOA based on Conditional Value at Risk

3.1 Original QAOA

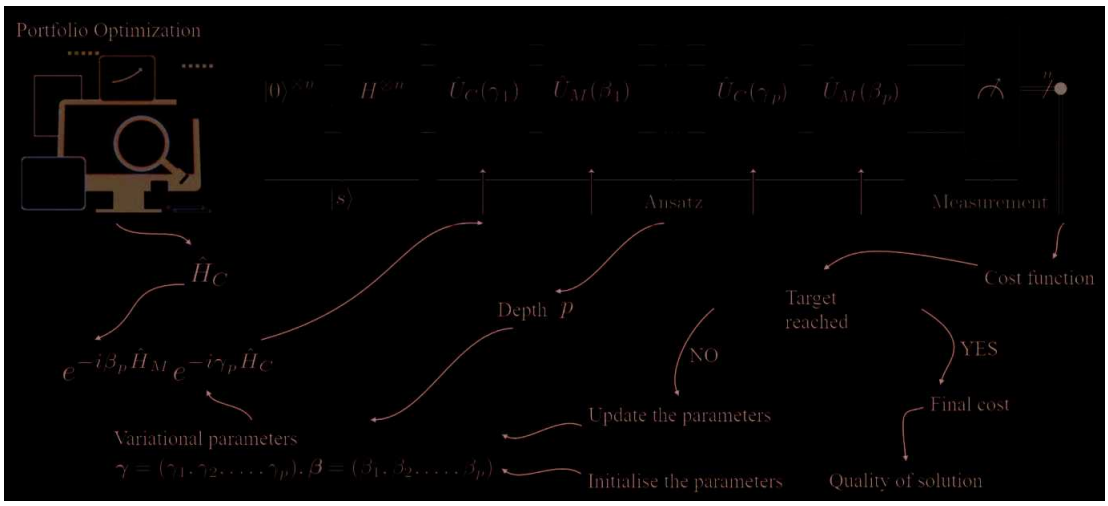

Fig. 2 General framework of original QAOA

QAOA serves as a heuristic method designed to approximate solutions to combinatorial optimization problems, operational on gate-based quantum computers. This section offers a succinct overview of the algorithm's original framework. The cost function $\mathcal{C}(\boldsymbol{z})$ can be formulated as a Hamiltonian $\hat{H}_C$ such that $\hat{H}_C|\boldsymbol{z}\rangle = \mathcal{C}(\boldsymbol{z}|\boldsymbol{z}\rangle$. The QAOA consists of the following 6 steps:

(1) Establish the Hamiltonians for both the cost and the mixing operations, as shown in Figure 2 procedure ①.

Let's reconsider Equation 3, which represents the quadratic Ising model. The model's structure indicates that $\mathcal{C}(\boldsymbol{z})$ consists of a sum of m individual terms, with each term involving a maximum of two spins.$\mathcal{C}(\boldsymbol{z}) = \sum_{k=1}^{m} \mathcal{C}^k(\boldsymbol{z})$ Equation 3 can be reformulated as a Hamiltonian for a system comprising n qubits: this is achieved by substituting z_i with the Pauli Z operator acting on the i_{th} qubit. Additionally, each interaction term $z_i z_j$ in equation 3 is reformulated as $\sigma_Z^i \otimes \sigma_Z^j$. Moreover, since every spin variable z_i corresponds to an eigenvalue of the Pauli-Z operator σ_Z^i, $\mathcal{C}(\boldsymbol{z})$ can be expressed in an operator format, $\hat{H}_C$, we can then define a unitary transformation that is parameterized.

$$\hat{H}_C = \sum_{i=1}^{n} d_i \sigma_Z^i + \sum_{i,j=1}^{n} Q_{ij} \sigma_Z^i \otimes \sigma_Z^j \tag{4}$$

Equation 4 is commonly referred to as the cost unitary or the phase separation unitary. Subsequently, we introduce a straightforward mixing operator, specifically an X-mixer.

$$\hat{H}_M = \sum_{j=0}^{n-1} \sigma_X^j$$

(2) Set the initial state of the circuit to $|s\rangle$, as shown in Figure 2 procedure ②. The system's states are established in a uniform superposition.

$$|s\rangle = |+\rangle^{\otimes n} = \frac{1}{\sqrt{2^n}} \sum_{\boldsymbol{x} \in \{0,1\}^n} |\boldsymbol{x}\rangle$$

where the summation encompasses all possible arrangements of the n-bit basis state $|\boldsymbol{x}\rangle$ (here, $\boldsymbol{x}$ represents the bit string format of $x_0 x_1 \cdots x_{n-1}$ pertaining to the bit vector $\boldsymbol{x}$.

(3) Assemble the quantum circuit design by specifying and implementing the unitary transformations, as shown in Figure 2 procedure ③. The method then continues by deploying a unitary operator consisting of p layers onto $|s\rangle$ to derive a quantum state with tunable parameters $|\psi(\gamma, \beta)\rangle$

$$U_C(\gamma) = e^{-i\gamma(\sum_{i=1}^{n} d_i \sigma_Z^i + \sum_{i,j=1}^{n} Q_{ij} \sigma_Z^i \otimes \sigma_Z^j)}$$
$$U_M(\beta) = e^{-i\beta(\sum_{i=1}^{n} d_i \sigma_X^i)}$$

This yields the trial wave function:

$$|\psi_p(\gamma, \beta)\rangle = e^{-i\beta_p \hat{H}_M} e^{-i\gamma_p \hat{H}_C} \ldots e^{-i\beta_1 \hat{H}_M} e^{-i\gamma_1 \hat{H}_C} |s\rangle$$

where γ and β are variational parameters within the circuit. We call $U_C(\gamma)$ and $U_M(\beta)$ the cost and mixer layers, respectively. A layer of QAOA includes a single cost layer and a single mixer layer, that can be sequentially layered to construct a more complex circuit with additional layers. Each mixer layer may be enacted using a solitary R_X

rotation gate. However, the two-qubit Pauli-Z interactions within the cost layer are realized by positioning a local rotation gate R_Z between two CNOT gates.

(4) Specify QAOA layers p and initialization of the $2p$ parameters, as shown in Figure 2 procedure ④.

Set the initial values for the $2p$ variational parameters $\gamma = (\gamma_1, \gamma_2, \ldots, \gamma_p)$,and $\beta = (\beta_1, \beta_2, \ldots, \beta_p)$ such that $\gamma_k \in [0, 2\pi)$ and $\beta_k \in [0, \pi)$ for $k = 1, \ldots, p$. Consequently, the ultimate state emitted from the circuit can be described as

$$|\psi_p(\gamma, \beta)\rangle = e^{-i\beta_p \hat{H}_M} e^{-i\gamma_p \hat{H}_C} \ldots e^{-i\beta_1 \hat{H}_M} e^{-i\gamma_1 \hat{H}_C} |s\rangle$$

(5) Definition of the Cost function, as shown in Figure 2 procedure ⑤.

The expectation value of the Hamiltonian $\hat{H}_C$ based on the ansatz state $|\psi_p(\gamma, \beta)\rangle$, which related to the cost associated with the quantum algorithm for the MVPO problem is determined by repeatedly measuring the final state:

$$F_p(\gamma, \beta) = \langle \psi_p(\gamma, \beta) | \hat{H}_C | \psi_p(\gamma, \beta) \rangle$$

In our method, the object is calculated based on CVaR which will be discussed in section 3.2

(6) Parameters optimization, as shown in Figure 2 procedure ⑥.

An iterative classical optimization technique is applied to refine the parameters γ and β . The aim of the previously mentioned process is to determine the best set of parameters (γ^*, β^*) to the effect that the expectation value $F_p(\gamma, \beta)$ is minimized: $(\gamma^*, \beta^*) = arg \min_{\gamma, \beta} F_p(\gamma, \beta)$.

3.2 Improved QAOA based on CVaR

QAOA employs a parameterized quantum circuit $U(\gamma, \beta)$ to construct candidate wave functions $|\psi(\gamma, \beta)\rangle = U(\gamma, \beta)|0\rangle$, utilizing a classical optimization algorithm that seeks to address the following objective:

$$\min_{\gamma, \beta} \langle \psi(\gamma, \beta) | H | \psi(\gamma, \beta) \rangle \tag{5}$$

It gets the total energy of a system by utilizing its Hamiltonian H, and its expected value $\langle \psi(\gamma, \beta) | H | \psi(\gamma, \beta) \rangle$. In sections 2 and 3.1, we have mentioned how to translates combinatorial optimization problems into the realm of Hamiltonians.

3.2.1 Optimization based on CVaR

The classical optimization algorithm must compute the expected value of objective function (5). Each observed string can be mapped to a sample from the expected value $\langle \psi(\gamma, \beta) | H | \psi(\gamma, \beta) \rangle$ since H is a weighted sum of tensor products of Pauli Z-matrices. The resulting samples are denoted by $H_k(\gamma, \beta)$, $k = 1, \ldots, K$, where $K \in \mathbb{N}$ represents the number of samples. Mean of the sample data $\frac{1}{K} \sum_{k=1}^{K} H_k(\gamma, \beta)$ provides an estimator for $\langle \psi(\gamma, \beta) | H | \psi(\gamma, \beta) \rangle$, and serves as the target function for the classical optimization algorithm.

To enhance the smoothness of the objective function while maintaining a focus on optimizing the best-measured outcomes instead of the average, we suggest employing the CVaR as the target function. CVaR represents the expected value of the lower

α-tail of the distribution of X. Without affecting generality, let us assume that the samples H_k are arranged in non-decreasing order, i.e., $H_{k+1} \geq H_k$. Consequently, for a given set of samples $\{H_k\}_{k=1,\ldots,K}$ and a confidence level $\alpha \in (0, 1]$, the $CVaR_\alpha$ is defined as follows:

$$CVaR_\alpha(X) = \lceil \alpha K \rceil^{-1} \sum_{k=0}^{\lceil \alpha K \rceil} H_k(\gamma, \beta)$$

The limit as α approaches zero corresponds to the minimum value, while $\alpha = 1$ corresponds to the expected value of X. In this context, CVaR serves as a generalization that encompasses both the sample mean and the best observed sample $\min\{H_1, \ldots, H_K\}$. For small, non-zero values of α, CVaR retains a focus on the best observed samples while also providing a smoother and more manageable objective function. By simply substituting the sample mean with $CVaR_\alpha$ in the classical optimization algorithm, we obtain the CVaR-QAOA algorithm. So the 6th step of section 3.1 is modified to find the optimal set of parameters (γ^*, β^*) such that the $CVaR_\alpha$ is minimized.

3.2.2 QAOA Ansatz variant

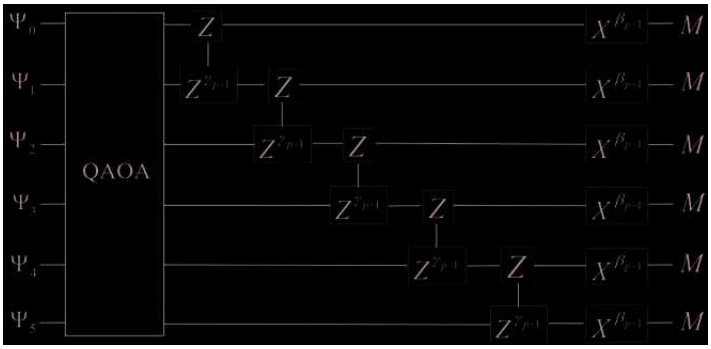

Fig. 3 The circuit layout for the Improved QAOA comprises a standard p QAOA layer, augmented by a problem-agnostic layer that incorporates ZZ gates, and a supplementary mixer layer utilizing X gates. Notably, the ZZ gates share the same parameter β, and similarly, the X gates also share an identical γ parameter .

We present an enhanced QAOA ansatz comprising a standard p QAOA layer augmented with a problem-agnostic circuit ansatz (depicted in Fig 3). Specifically, the phase separation layer connects qubits in a sequential manner from the first to the last. This additional layer employs a minimal set of n-1 two-qubit gates that minimally entangle all qubits. We employ a single phase for all two-qubit ZZ gates and another single mixing layer phase for all single-qubit X gates.This improved ansatz offers a favorable balance between the performance of $p = 1$ and $p = 2$ QAOA for portfolio optimization problems, while maintaining a relatively shallow QAOA circuit depth.

The final state of this ansatz is:

$$|\psi'_p(\gamma, \beta)\rangle = e^{-i\beta_{p+1}\hat{H}_M} e^{-i\gamma_{p+1}\sum_{j=1}^{n-1}\sigma_Z^j \otimes \sigma_Z^{j+1}} e^{-i\beta_p \hat{H}_M} e^{-i\gamma_p \hat{H}_C} \ldots e^{-i\beta_1 \hat{H}_M} e^{-i\gamma_1 \hat{H}_C} |s\rangle$$

However, in section 3.1 (4), we need to initialize the $2(p+1)$ variational parameters $\gamma = (\gamma_1, \gamma_2, \ldots, \gamma_p, \gamma_{p+1})$ and $\beta = (\beta_1, \beta_2, \ldots, \beta_p, \beta_{p+1})$ such that $\gamma_k \in [0, 2\pi)$ and $\beta_k \in [0, \pi)$ for $k = 1, \ldots, p+1$.

4 Experiments

Table 1 Company name of 16 stocks

No.	Company Name of Stocks
1	Alibaba Group Holding Limited American Depositary Shares each representing eight Ordinary share (BABA)
2	Anta Sports Products (ANPDF)
3	Baidu, Inc. ADS (BIDU)
4	JD.com, Inc. American Depositary Shares (JD)
5	China Petroleum & Chemical Corp - Class H (SNPMF)
6	Amazon.com, Inc. Common Stock (AMZN)
7	Hitachi Ltd. ADR (HTHIY)
8	Mitsubishi Corporation (MSBHF)
9	Siemens AG ADR (SIEGY)
10	Tesla, Inc. Common Stock (TSLA)
11	Apple Inc. Common Stock (AAPL)
12	GE Aerospace Common Stock
13	Microsoft Corporation Common Stock (MSFT)
14	QUALCOMM Incorporated Common Stock (QCOM)
15	Nike, Inc. Common Stock (NKE)
16	Adidas AG ADR - Level I (ADDYY)

We generate portfolio instances for 10, 12, 14, and 16 qubits. For each instance, we employ the CVaR objective function with α values ranging from 1% to 100% and test QAOA with $p = 1$ and $p = 2$. However, the experimental results for $p = 2$ do not outperform $p = 1$, hence we only present the results for $p = 1$ in figure 4. To optimize our portfolios, we download historical stock data from Nasdaq spanning five years from 04/22/2019 to 04/18/2024. Table 1 lists the 16 prominent stocks selected for our experiments.

Based on the approach described in [29], we utilize the classical optimizer COBYLA to optimize the parameters of the trial wave function. For our experimental evaluation, we assess the performance of various algorithms using the exact quantum state simulations provided by TensorCircuit, an open-source quantum circuit simulator [30]. This enables us to accurately evaluate and characterize the performance metrics we employ.

CVaR-QAOA and Improved CVaR-QAOA, each with seven different values of $\alpha = (0.01, 0.05,\ 0.1, 0.25,\ 0.5, 0.75, 1)$, will be executed, and a callback function will record the parameters throughout the optimization process. When $\alpha = 1$, the full measurement results are accepted, and the model reverts to the standard QAOA. The figures demonstrate that using CVaR leads to faster convergence towards a lower loss. The data indicates that QAOA with CVaR typically exhibits a higher probability

of achieving the desired measurement outcome. As shown in Fig. 4, Fig. 5, Fig. 6, and Fig. 7, the improved CVaR-QAOA achieves a lower cost value faster than the CVaR-QAOA regardless of the number of stocks considered. As depicted in Fig. 4

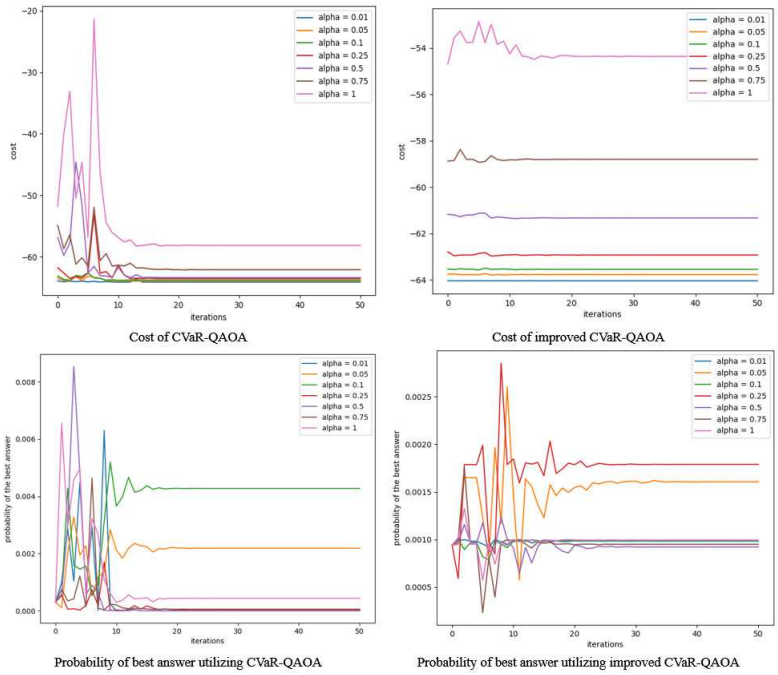

Fig. 4 Comparisons of CVaR-QAOA and our method with 10 Stocks, Budget = 7 and p=1

for the 10-stock and B=7 scenario, our method exhibits superior performance when $\alpha \in [0.01, 0.05]$, while the CVaR-QAOA achieves its best results with $\alpha = 0.1$. As illustrated in Fig. 5 for the 12-stock case, our method achieves optimal performance with $\alpha = 0.05$, while the CVaR-QAOA performs best when $\alpha = 0.01$. As shown in Fig. 6 for the 14-stock scenario, our method exhibits superior performance when $\alpha = 0.1$, while the CVaR-QAOA achieves optimal results with $\alpha = 0.01$. As depicted in Fig. 7 for the 16-stock case, our method achieves the best performance when $\alpha = 0.01$, whereas the CVaR-QAOA shows optimal results with $\alpha = 0.1$. Notably, our method significantly outperforms the CVaR-QAOA, exhibiting a much faster cost reduction and a higher probability of success when $\alpha = 0.01$. These results indicate that as the number of stocks increases, our method's advantages become more pronounced. The figures and Table 2 illustrate that the utilization of CVaR leads to faster convergence towards a lower loss than the original QAOA ($\alpha = 1$). In most cases, $\alpha = 0.01$ performs better than other values in portfolio optimization problem. When the stocks num is 16, the original QAOA needs 700 iterations to get the optimal value, however our improved CVaR-QAOA only still less than 40 iterations with $\alpha = 0.01$. The data indicates that QAOA with CVaR typically exhibits a higher likelihood of obtaining the correct measurement outcome. Furthermore, our method performs optimally when

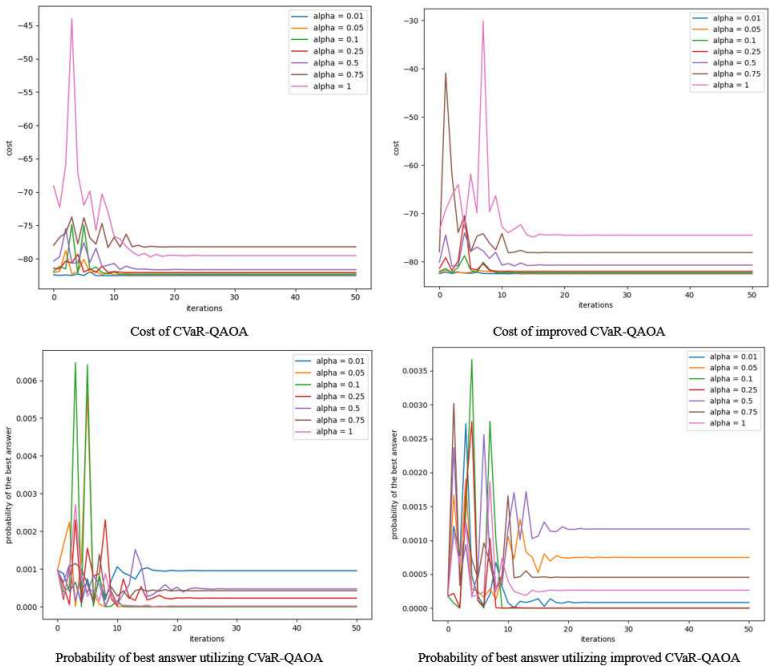

Fig. 5 Comparisons of CVaR-QAOA and our method with 12 Stocks, Budget=8 and p=1

Table 2 Iterations required for convergence to the optimal cost value

iterations	10Stocks	12Stocks	14Stocks	16Stocks
original QAOA	500	550	600	700
CVaR-QAOA ($\alpha = 0.1$)	100	50	150	200
our method ($\alpha = 0.1$)	100	30	100	100
CVaR-QAOA ($\alpha = 0.01$)	40	30	30	50
our method ($\alpha = 0.01$)	30	30	25	35
optimal value	-64.1406	-82.6233	-126.593	-126.9287

$\alpha \in [0.01, 0.1]$. Our improved CVaR-QAOA demonstrates even faster convergence compared to the standard CVaR-QAOA and achieves a higher probability of finding the best answers, particularly when portfolio optimization involves a larger number of stocks.

5 Conclusion

The proposed improved QAOA leverages CVaR as a risk metric to focus on the tail end of the loss distribution, leading to a more robust optimization approach. The improved

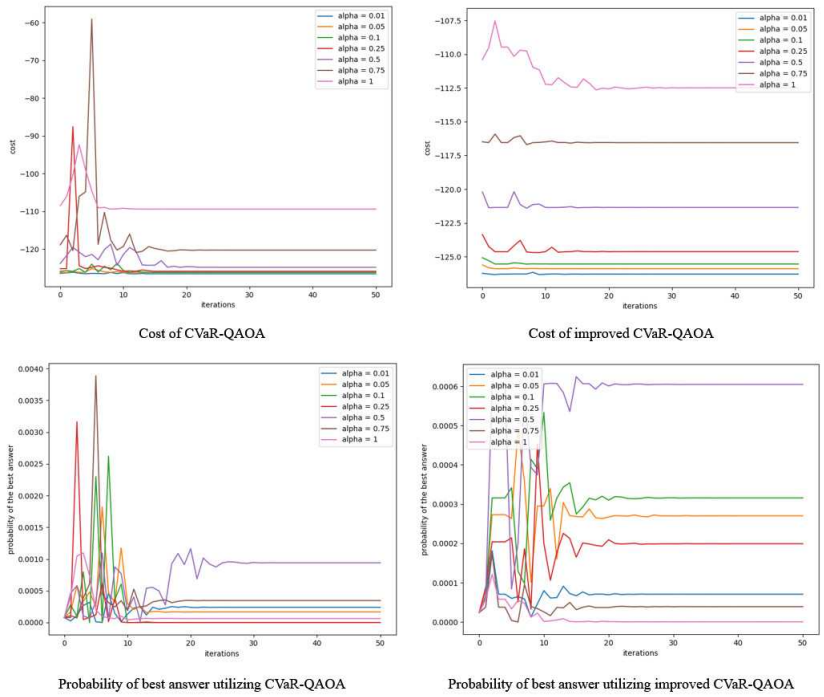

Fig. 6 Comparisons of CVaR-QAOA and our method with 14 Stocks, Budget=10 and p=1

QAOA ansatz we proposed includes an additional problem-independent circuit layer, aiming to achieve a balance between optimization quality and circuit depth. Experiments results indicate that the improved QAOA with CVaR converges more rapidly to lower loss values and has a higher probability of obtaining the correct measurement outcome, especially for larger portfolios.This paper concludes that the improved QAOA with CVaR is a promising approach for portfolio optimization, particularly as the number of assets increases, demonstrating its potential for practical applications in the financial industry. However, we are not sure which definite α value can obtain the best performance. Whether there is a self-adaptive approach to choose the best parameter for α, so that different portfolio optimization problems can work much more effectively. The future research direction is to find the model with high-performance and stable without being overly reliant on or sensitive to parameter adjustments.

Supplementary information. All authors contributed to the study conception and design. Material preparation, data collection and analysis were performed by Rong Jin. The first draft of the manuscript was written by Qingqing Yu and all authors commented on previous versions of the manuscript. All authors read and approved the final manuscript.

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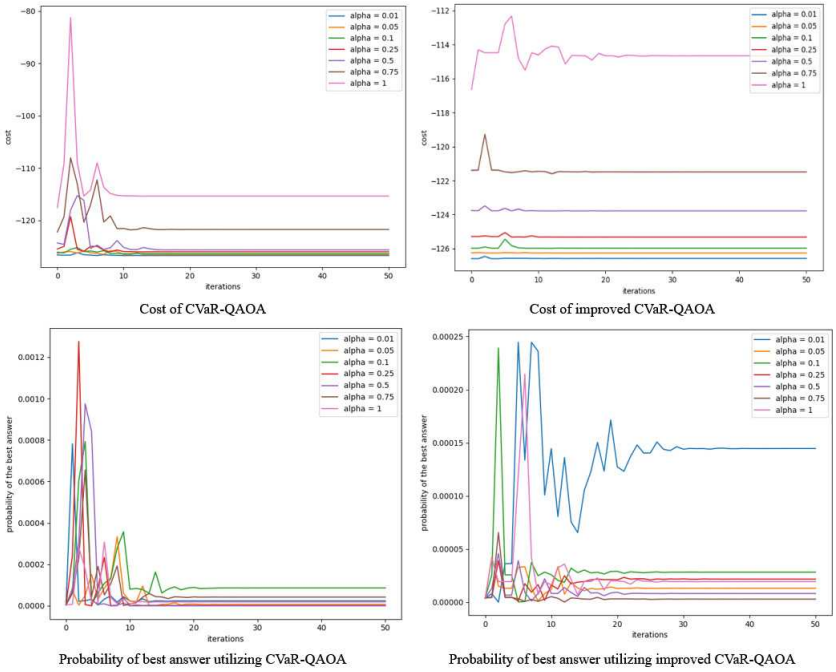

Fig. 7 Comparisons of CVaR-QAOA and our method with 16 Stocks, Budget = 10 and p=1

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